Jiawei Liu

Curriculum Vitae

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Education

Jan. 2021 – Present **National University of Singapore**, *Doctor of Philosophy*, Institute of Data Science, NUS Graduate School for Integrative Sciences and Engineering, Research topic: 3D avatar, video generation. Main Advisor: Prof. Mike Zheng Shou, Co-advisor: Prof. Jussi Keppo.

Sept. 2018 – July 2020 **Tsinghua University**, *Master of Science*, Department of Engineering Mechanics, Institute of Dynamics and Control, Advisor: Prof. Gexue Ren, Cumulative GPA: 3.70/4.0.

Sept. 2014 – June 2018 **Northwestern Polytechnical University**, *Bachelor of Engineering*, **Honors College**. Cumulative GPA: 3.80/4.0, 91.84/100, Rank: 2/145, Major GPA: 3.88/4.0, 93.96/100.

Standardized Test

May 2019 TOEFL iBT: R28/L30/S21/W27, Total: 106.

Jan. 2019 GRE General: Verbal: 158(80%), Quantitative: 169(95%), AW: 4.0(57%), Total: 327+4.0.

Publications

Aug. 2019 **Jia-Wei Liu**, Jia-Peng Liu, Xuan-Bo Shu, Aki Mikkola, Ge-Xue Ren, "An efficient multibody dynamic model of three-dimensional meshing contacts in helical gear-shaft system and its solution", Mechanism and Machine Theory 142 (2019) 103607.

Feb. 2019 **Jia-Wei Liu**, Jia-Peng Liu, Jia-Lun Yao, Ge-Xue Ren, "Dynamic characteristics of helical gear based on Arbitrary Lagrangian Eulerian formulation", in: ECCOMAS 2019 Multibody Dynamic Conference, Duisburg, Germany, 2019.

Mar. 2020 Jiang Cui, Zhi-Hua Zhao, **Jia-Wei Liu**, Xuan-Bo Shu, Ge-Xue Ren, "Uncertainty analysis of mechanical dynamic using HHT and LMD", in: 25th International Congress of Theoretical and Applied Mechanics (25th ICTAM), 2020.

Sep. 2021 Jiang Cui, Zhi-Hua Zhao, **Jia-Wei Liu**, Peng-Xiang Hu, Ru-Neng Zhou, Ge-Xue Ren, "Uncertainty analysis of mechanical dynamics by combining response surface method with signal decomposition technique", Mechanical Systems and Signal Processing 158 (2021) 107570.

Research Interests

Computer vision, Autonomous vehicles, AI Robotics, Dynamics and control.

Research Experience

Jan. 2021 – Nov. 2021 **Combining 3D geometry with deep learning for robust registration and retrieval**, Research leader, National University of Singapore.

Bulit a multi-view line detection dataset based on MegaDepth dataset. Proposed a weakly-supervised line detection framework using trifocal tensor based multi-view consistency. Proposed a domain adaptive line detection and description framework using homography warping consistency.

Oct. 2018 – July 2020 **Model order reduction method for simulation of planetary gear set based on Arbitrary Lagrangian Eulerian (ALE) formulation**, Research leader, Tsinghua University.

Proposed a novel ALE planetary gear set to reduce the degrees of freedom (DOFs) of the system from 10^7 to 10^4. By splitting the planet gear into two ALE gears, each ALE planet gear only contacts with the ring or sun gear, making it feasible to develop ALE planetary gear set. Five times acceleration of simulation time with comparable simulation accuracy was achieved compared with ABAQUS.

Feb. 2019 – Aug. 2019 **An efficient multibody dynamic modeling method of three-dimensional (3D) contacts in helical gear-shaft system**, Research leader, Tsinghua University.

Proposed the ALE helical gear, where low-frequency approximation (LFA) technique and ALE formulation were adopted to reduce the system DOFs from 10^6 to $10^3 \sim 10^4$. Then, we simplified the inertial terms in dynamic equations and adopted a high-efficiency contact algorithm based on hierarchical oriented bounding box tree (OBB-tree). Finally, 3D Timoshenko beam elements were adopted to discretize the shaft. The proposed method was implemented using C++ and one order of magnitude acceleration of simulation time was achieved with high accuracy compared with ABAQUS.

June 2018 – Feb. 2019 **Development of evaluation method for sliding wear by multibody dynamics**, Project leader, Tsinghua University, Undertaken with: Mitsubishi Heavy Industries, Japan.

Established a general low-frequency approximated model using C++. Adopted the hierarchical OBB-tree method for rapid contact detection in variable-geometry turbocharger (VGT). Improved the computational efficiency for the dynamics simulation of VGT by five times.

Jan. 2018 – Oct. 2018 **Research on dynamic characteristics of helical gears and bevel gears based on ALE formulation**, Research leader, Tsinghua University.

Established the FE models of four pairs of helical gears and two pairs of bevel gears using Solidworks and Hypermesh. Developed the corresponding ALE gears for higher efficiency. Analyzed the dynamic transmission characteristics of gears, for which an algorithm to compute contact lines was proposed.

Work Experience

Sep. 2020 – Dec. 2020 **Computer Vision Internship**, Mech-Mind Robotics.

Deployed 3D vision projects, including selecting point cloud registration algorithms and deep learning models (ResNet, Mask R-CNN, DexiNed).

Jan. 2018 – July 2020 **Research Assistant**, Multibody Dynamics Laboratory, Tsinghua University.

Contributed to the research of efficient numerical algorithms in multibody dynamics, with applications on mechanical systems.

Sept. 2018 – Jan. 2019 **Teaching Assistant**, Theoretical Mechanics, Tsinghua University.

Provided 2 lectures on laboratory sessions. Offered homework correction twice a week and presented 5 Q&A sessions. Invigilated and graded on 3 exams.

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Volunteers & Activities

Sept. 2018 – Jan. 2019 **Principle Member**, League Graduate, Tsinghua University.

Assisted in the design of a memorial video and collaborated on encouraging students participation.

Sept. 2018 – Jan. 2019 **Principle Member**, Graduate Student Union, Tsinghua University.

Served as volunteer in the opening ceremony. Assisted in organizing academic lectures.

Apr. 2016 – June 2016 **Team Leader**, Project Management, Northwestern Polytechnical University.

Worked with teachers from Netherlands and led a team to organize the route plannings in Africa.

Sept. 2015 – Sept. 2016 **Principle Member**, Enactus Practice Center, Northwestern Polytechnical University.

Collected and analyzed data for the water-shortage region and collaborated on inventing our first water filter equipment, which adopted soil infiltration and was about 5 times cheaper than the market.

Sept. 2014 – July 2018 **Class Leader**, Honors College, Northwestern Polytechnical University.

Offered supports for students' study and organized dozens of Q&A sessions and study lectures.

Honors & Awards

Jan. 2021	DATA SCIENCE NGS SCHOLARSHIP, National University of Singapore (Awarded to 10 students every year).
June 2020	Outstanding graduate of School of Aerospace Engineering, Tsinghua University (Awarded to top 8% students).
Oct. 2019	Scholarship of Second Prize in Tsinghua University (Awarded to top 5% students).
Jan. 2018	Scholarship of Innovation and Entrepreneurship in Honors College.
June 2017	Second Prize of C++ Programming in Northwestern Polytechnical University.
Mar. 2017	Third Prize of National English Contest for College Student.
2016 – 2017	Scholarship of First Prize in Honors College.
2016 – 2017	Scholarship of Outstanding Student in Northwestern Polytechnical University.
May 2016	Second Prize of Mathematical Modeling in Northwestern Polytechnical University.
2015 – 2016	Scholarship of First Prize in Honors College.
2015 – 2016	Scholarship of Outstanding Student in Northwestern Polytechnical University.
May 2015	Second Prize of Mathematical Modeling in Northwestern Polytechnical University.
2014 – 2015	Scholarship of First Prize in Honors College.
2014 – 2015	Scholarship of Outstanding Student in Northwestern Polytechnical University.
2014 – 2015	National Scholarship (Awarded to top 1% students in Honors College).

Skills & Abilities

Proficient in	Python, C/C++ Programming, Matlab, Pytorch, OpenCV, ABAQUS, Hypermesh, Solidworks, Adobe Illustrator, Latex(Lyx), MS Office.